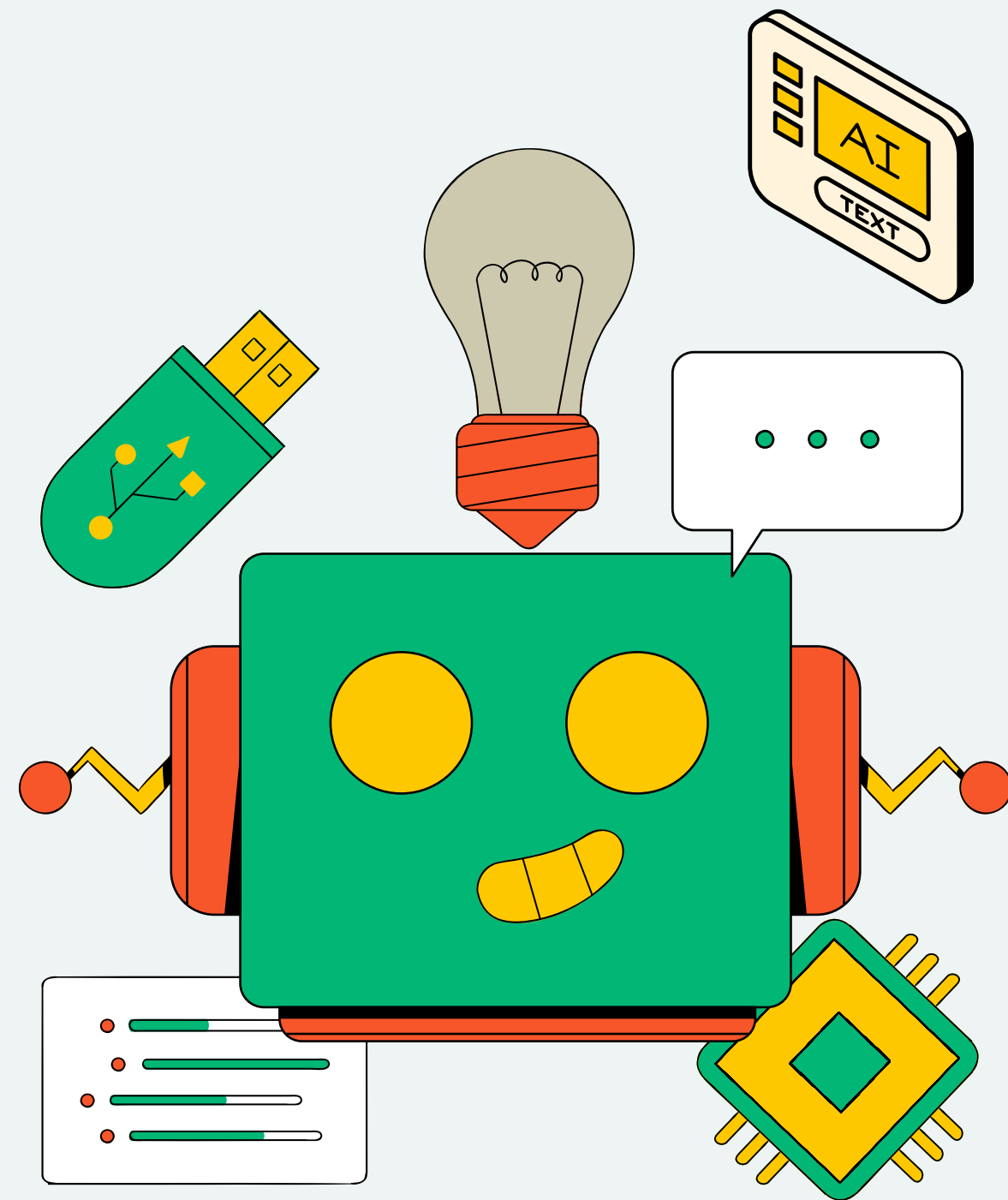
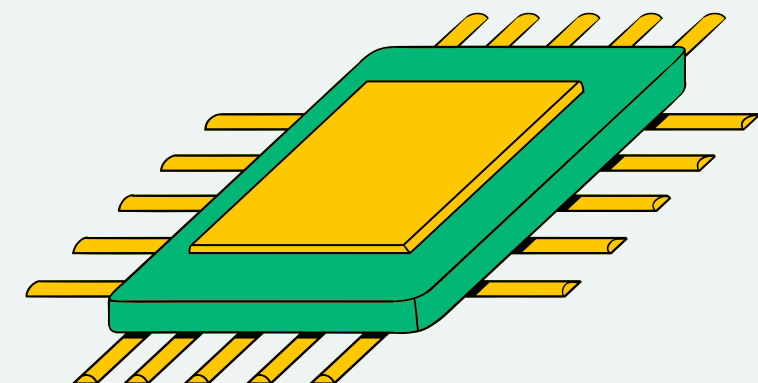




DATA VISUALIZATION PROJECT

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FLOW CREDIT

- 1.Information of dataset
- 2.Questions



DATA INFORMATION

General Description

- This company sells products (vehicles) with credit in Ecuador.
- The dataset contains records of credit-flow events for individual prospects that the company offers to its clients to buy the product.
- Each row represents a single prospect evaluation or credit-related event. (total = 26819)
- The data combines demographic information, risk assessment, approval workflow, and financial indicators.

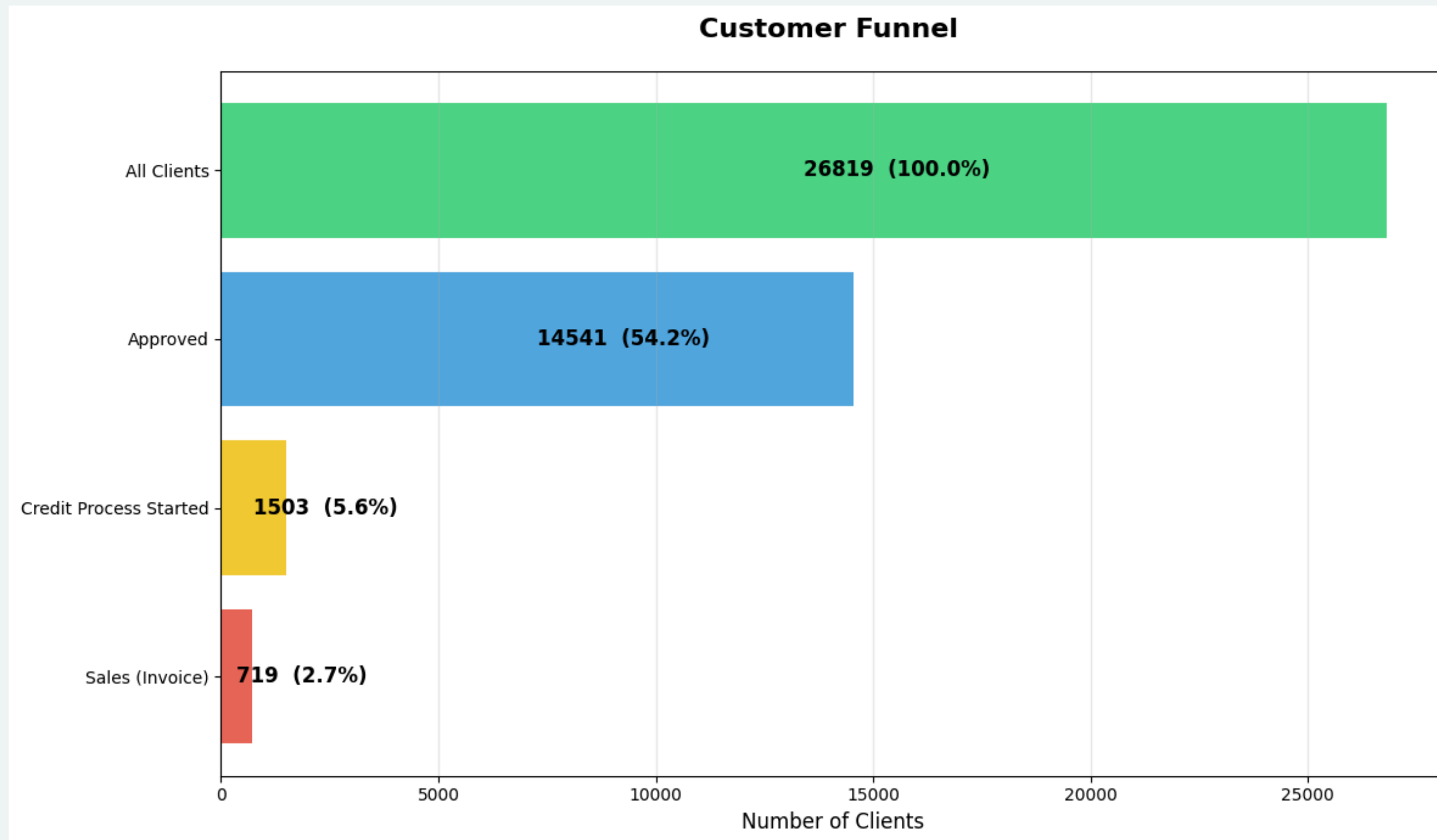
Key Characteristics

- Temporal coverage: 2025 from January to November
- Entity identifier: Each prospect is uniquely tracked through an anonymized id.
- Volume: Medium-sized dataset suitable for exploratory analysis, modeling, and dashboard.

DATA INFORMATION

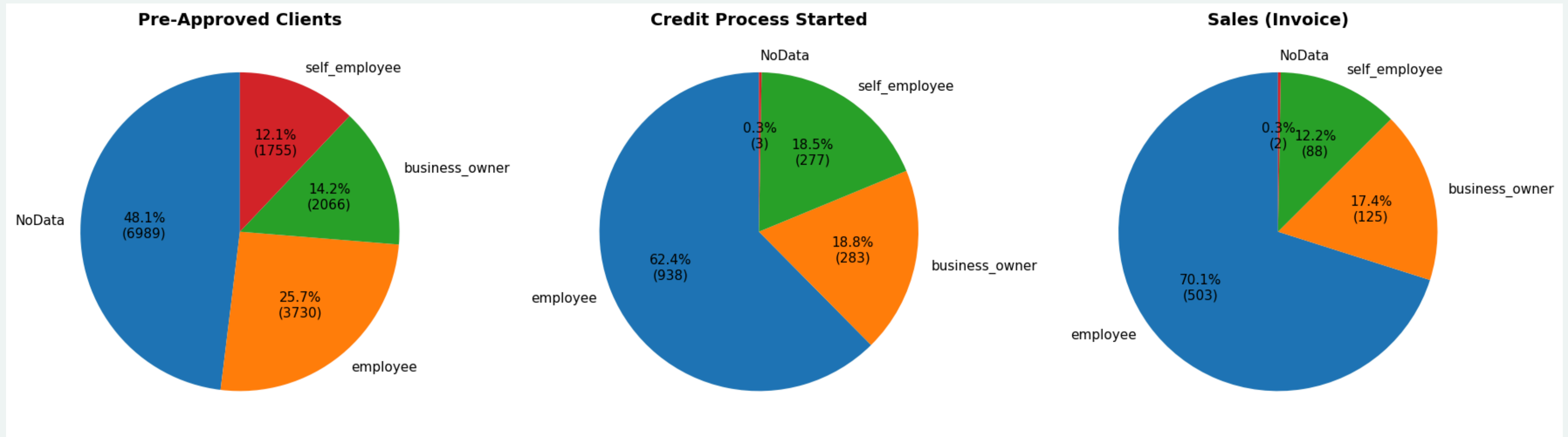
Column	Description
date	Timestamp of the credit event or record creation.
id	Unique anonymized identifier for each prospect.
age	Age of the prospect.
processed_prospect	Indicates whether the prospect has been processed in the workflow or rejected
approval_level_analyst	Analyst credit -level approval decision or status.
approval_level_automatic	Automatic system approval decision.
approval_level_supervisor	Supervisor credit -level approval decision.
billing	Indicates if the customer has invoice
event_income	Indicates if the credit process begin
risk_level	Numeric risk score.
formality	Indicates the formality level of the credit request (formal/informal/own business).
supervisor_sales	Supervisor responsible for the sales
province	Geographic location of the prospect.
income	Declared or verified income of the prospect.
financial_expenses	Reported financial expenses, used for affordability analysis.

QUESTION 1: WHAT IS THE FLOW OF THE CREDITS IN THE COMPANY?



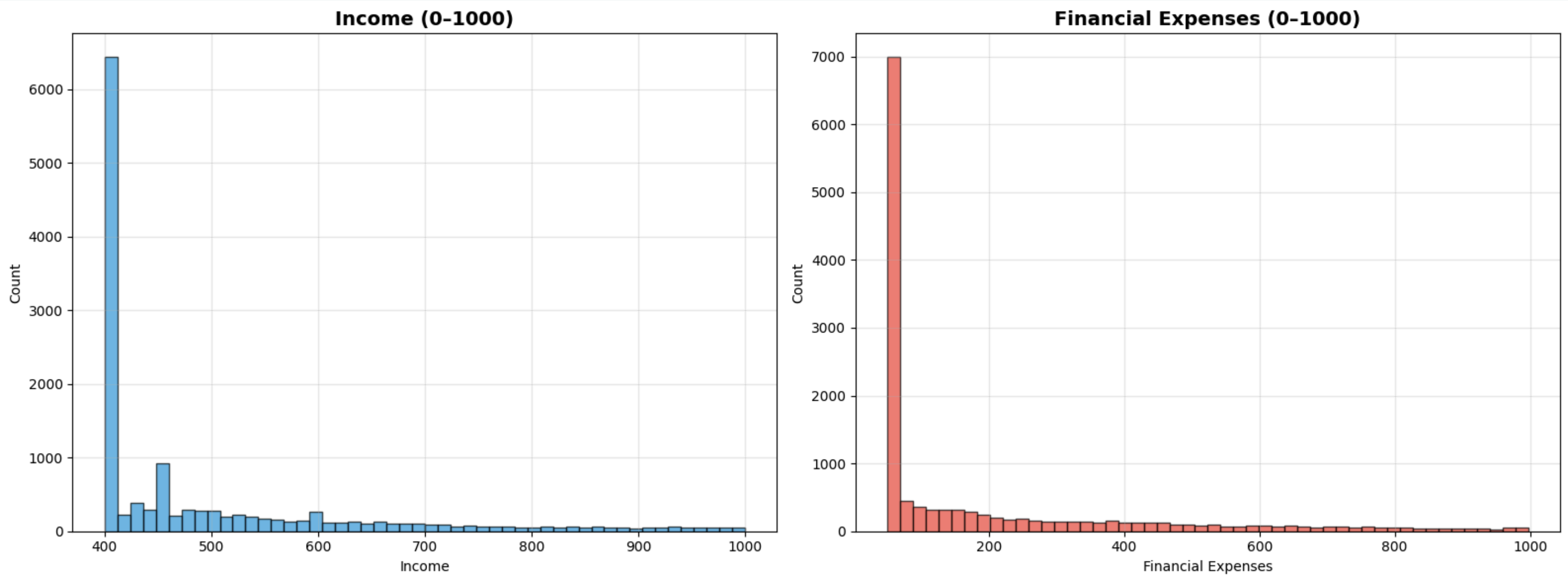
- In this period of time, we have a 2.7% of conversion rate of those interested in credit, which is a good outcome, but we can improve almost the 5.6%. We can analyze the reason why the credits that begin the process don't convert in sale.
- From the aproved we need to see if we can improve the score of the clients and get a bit of risk and star more credits. Almost get the 10% of the credit process started. Firt we need to see the reports of the people with default or people who don't pay the credit.
- Increase the marketing campaign for the credit to reach more people in order to increase the conversion.

QUESTION 2: WHAT IS THE DISTRIBUTION OF PRE-APPROVED, APPROVED, CREDIT STARTED, AND SALES WITH FORMALITY?



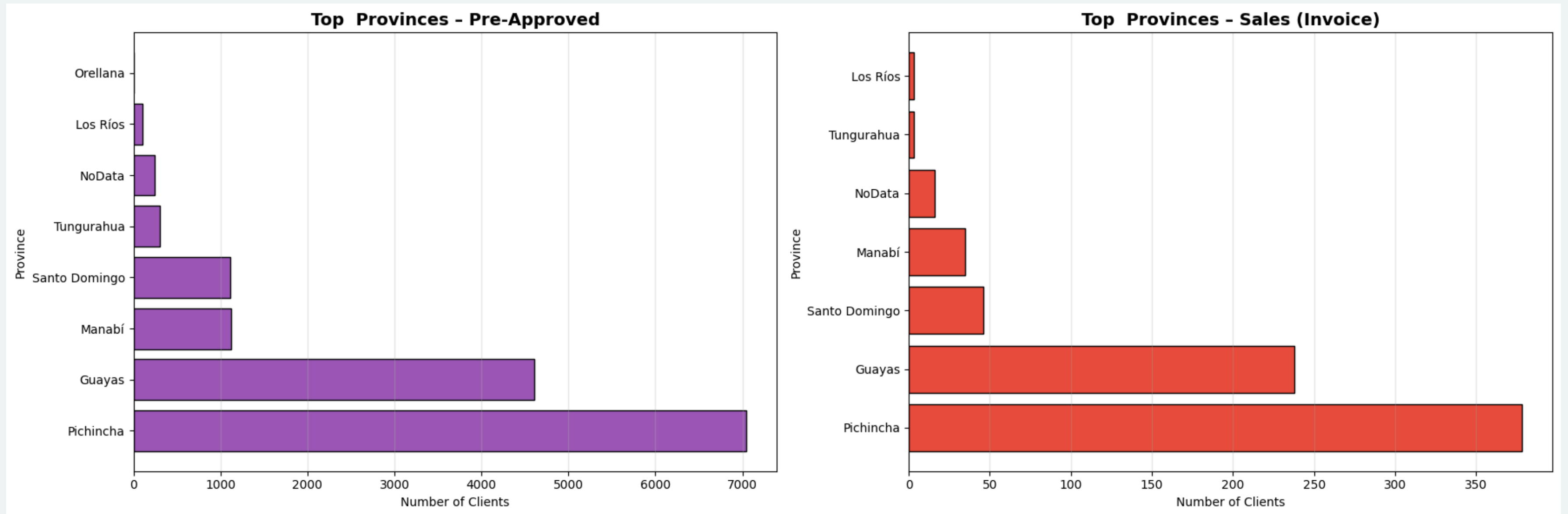
- We can notice that the people with formal employee has more credits, it is a good signal but we can improve until 80%.
- Also we know if the people doesn't have information about their employee is rejected, only we have some people with NO DATA that pass through the filter due to exceptions.

QUESTION 3: HOW DOES THE INCOME AND EXPENSE DISTRIBUTION LOOK ACROSS ALL CUSTOMERS WITH A CREDIT FORM?



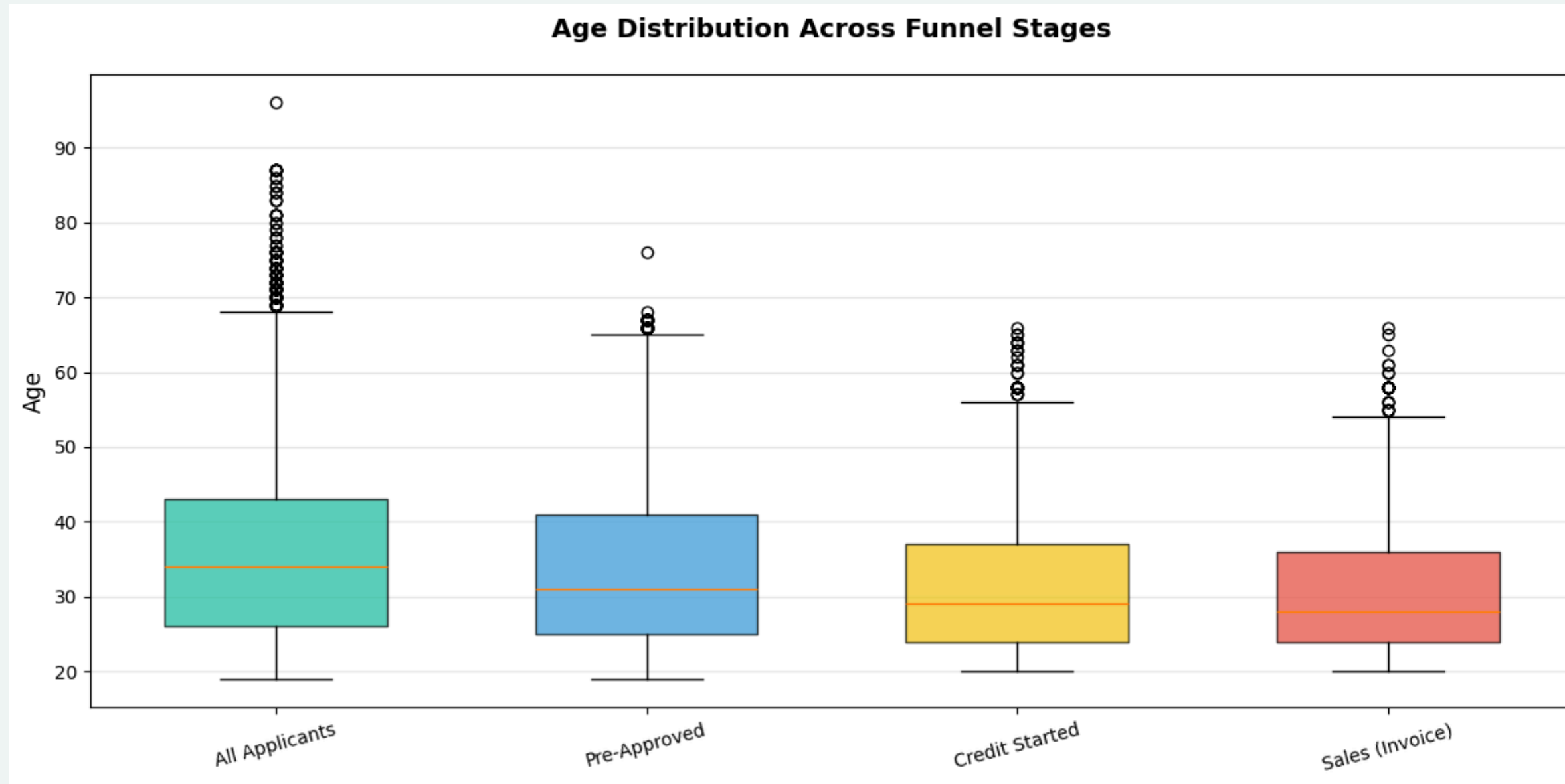
The majority of our clients have low income and also low expenses. Our segment of market is risk but is the field we accept the sales of this amount od credits.

QUESTION 4: WHAT ARE THE TOP PROVINCES BY CREDIT PRE-APPROVAL BY PROVINCE AND CONVERSION?



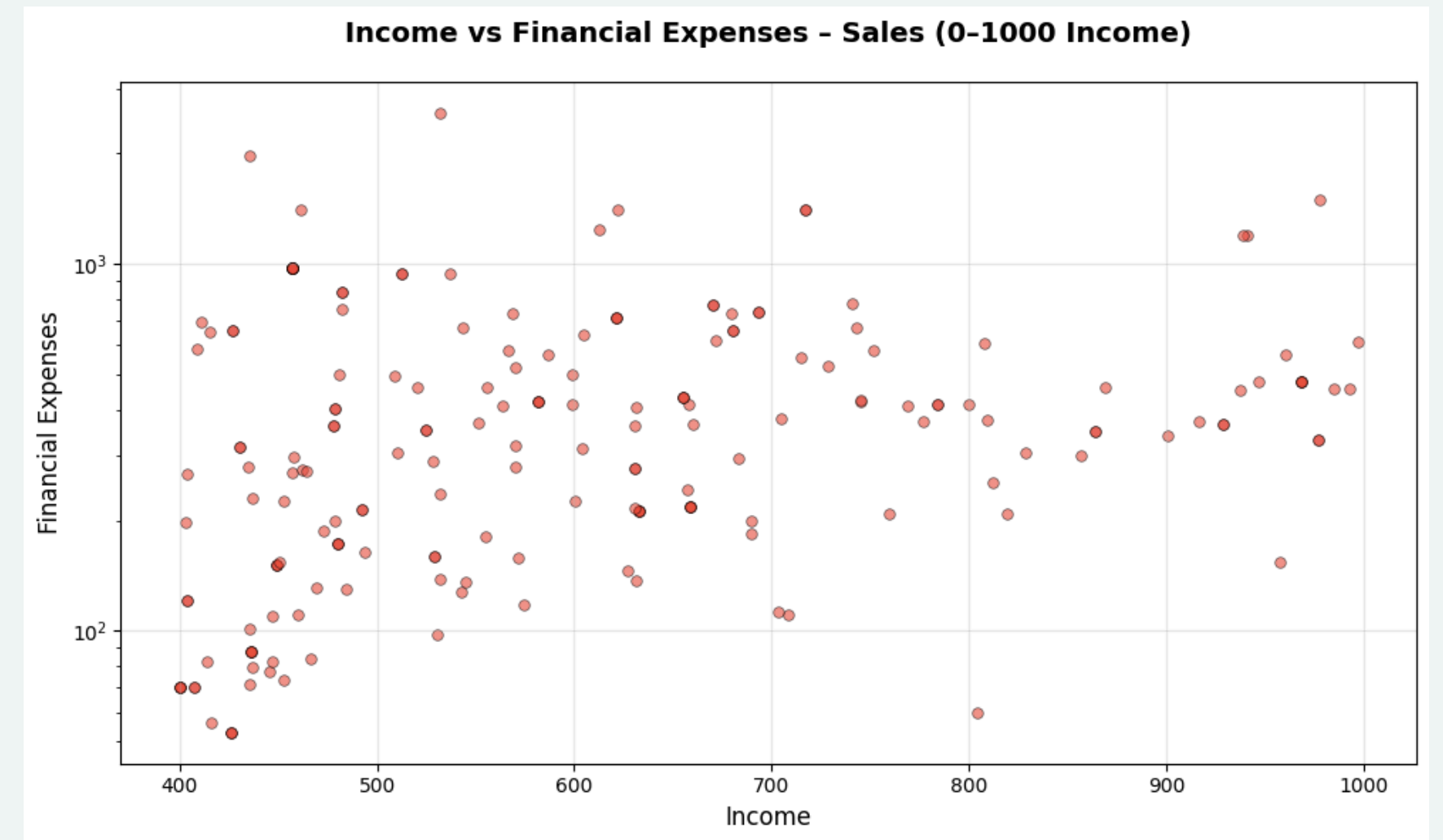
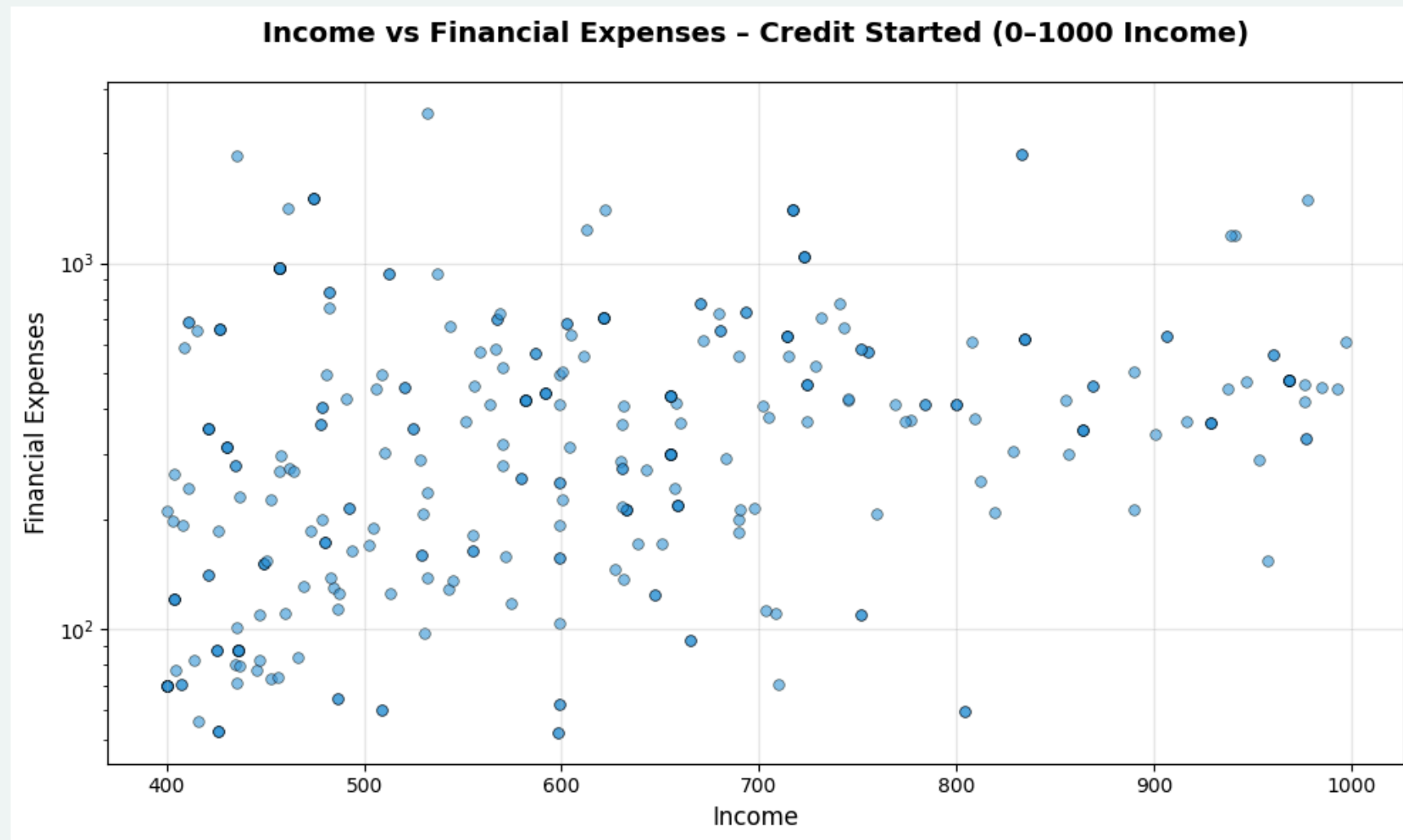
- We see the same pattern but with a slight difference, for example Santo Domingo has a better conversion, NoData info, are the clients from web, not a specific store.
- Finally, to improve the sales in Guayas, one of the largest cities in Ecuador, we can suggest a strong marketing campaign and also the possibility to create a new physical store.

QUESTION 5: WHAT IS THE AGE DISTRIBUTION OF APPLICANTS BY EACH STAGE?



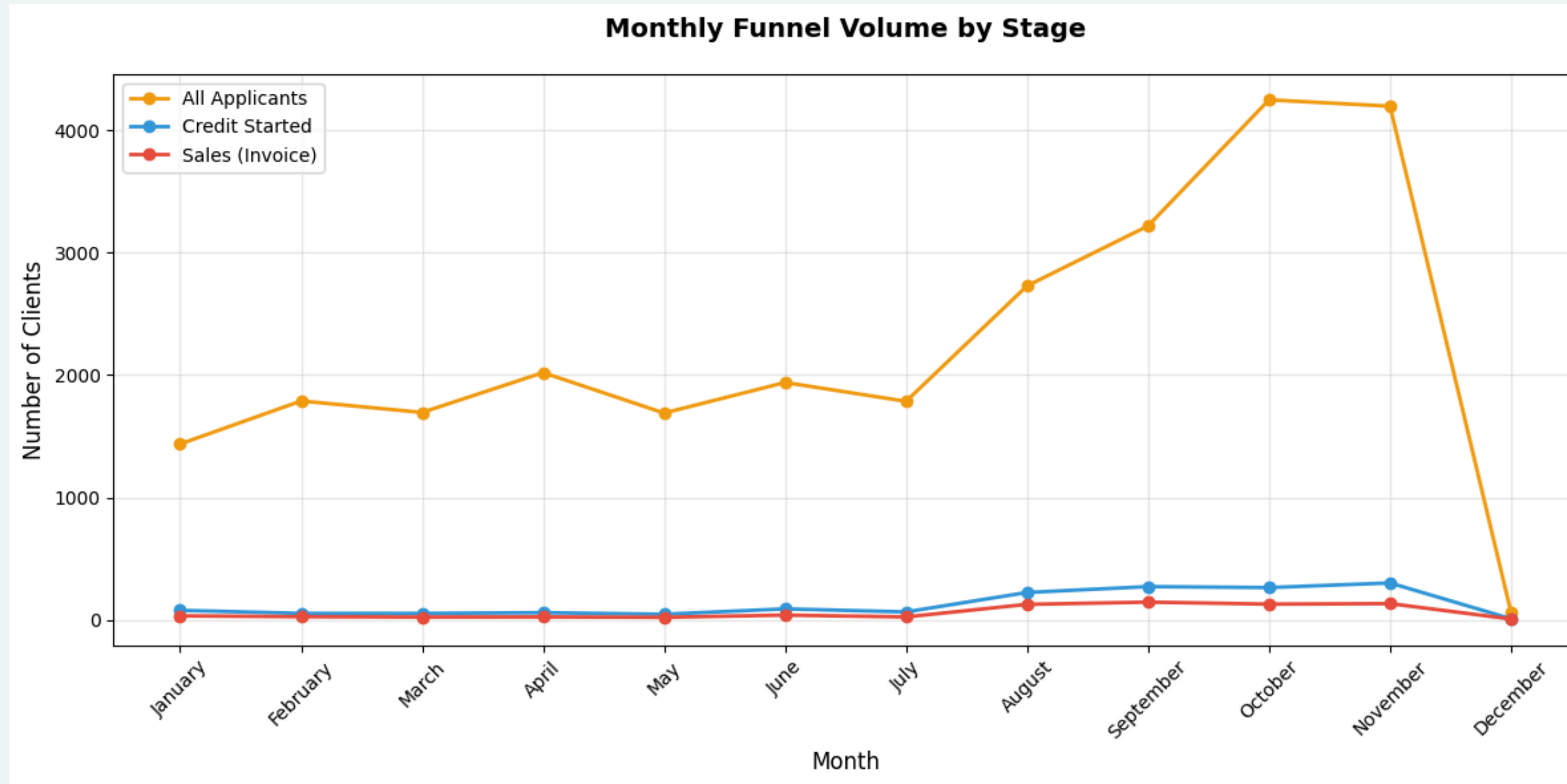
- The top of the funnel attracts people from a very wide age range, including many older applicants.
- As the funnel progresses, older ages drop out, and no one above 68 reaches the sales stage.
- The median age consistently decreases from applicants, pre-approved, credit started, and conversion.
- Your product converts best among younger clients, while older applicants show interest but rarely complete the process.

QUESTION 6: HOW DO FINANCIAL EXPENSES RELATE TO INCOME?



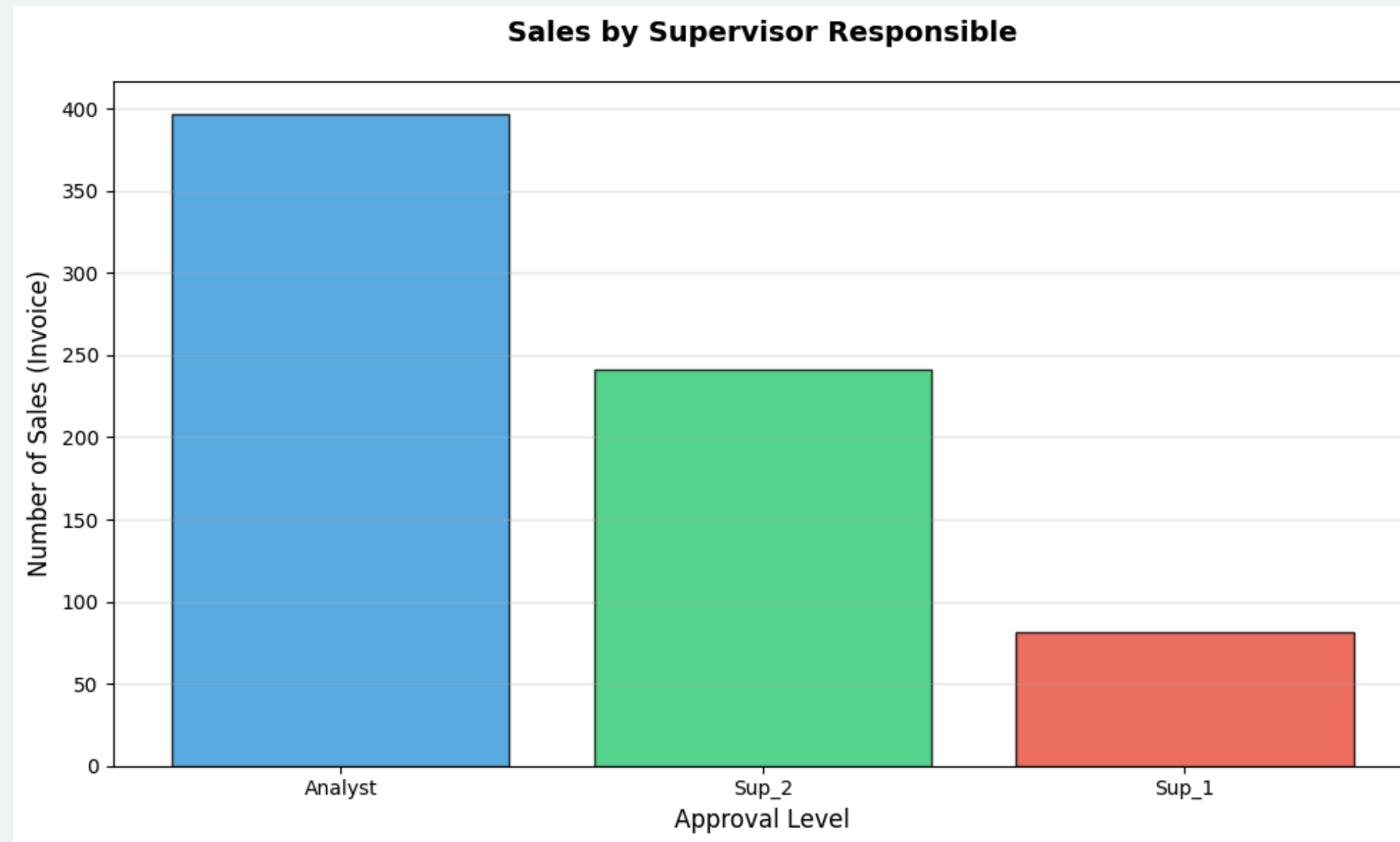
- Financial expenses still vary widely, but the log scale reveals a more structured pattern instead of a chaotic vertical cloud.
- Lower-income clients show greater variability in expenses, while higher-income clients cluster more tightly, suggesting more stable financial behavior.
- Finally, we can see the pattern that we can suggest, and it is that with more income we have more expenses.

QUESTION 7: HOW MANY CREDITS WERE PROCESSED EACH MONTH?



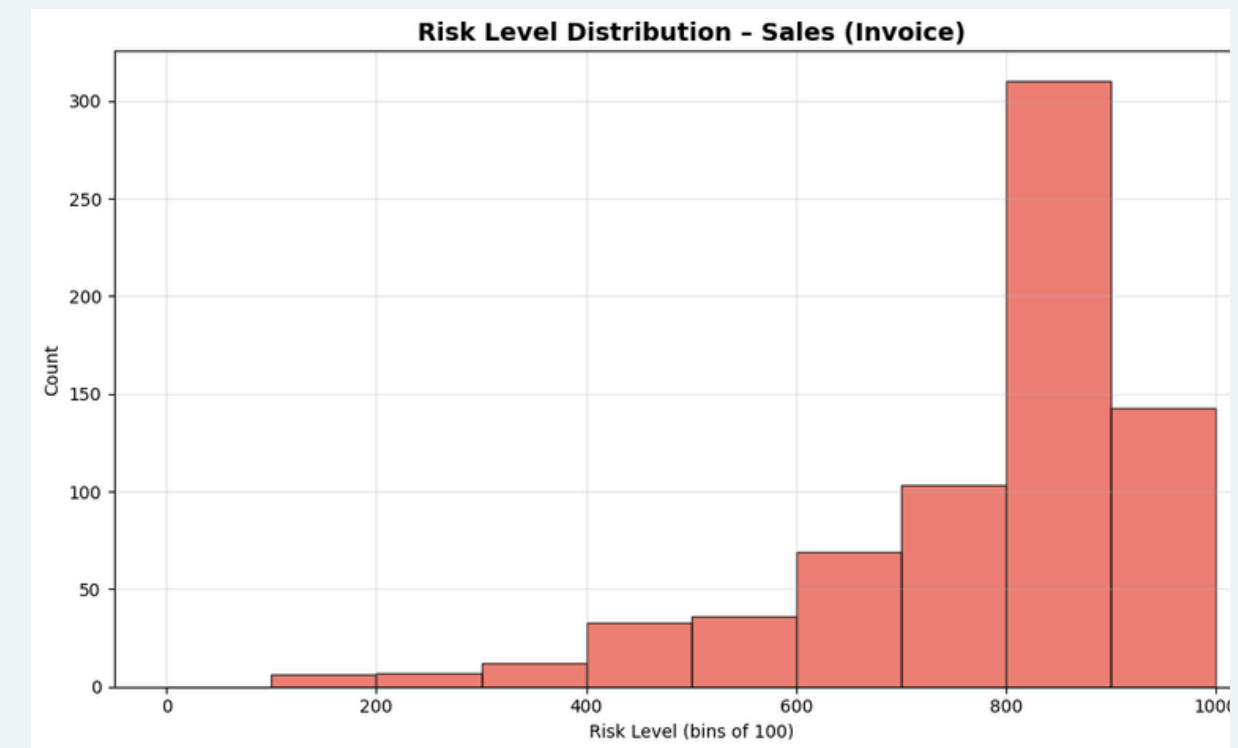
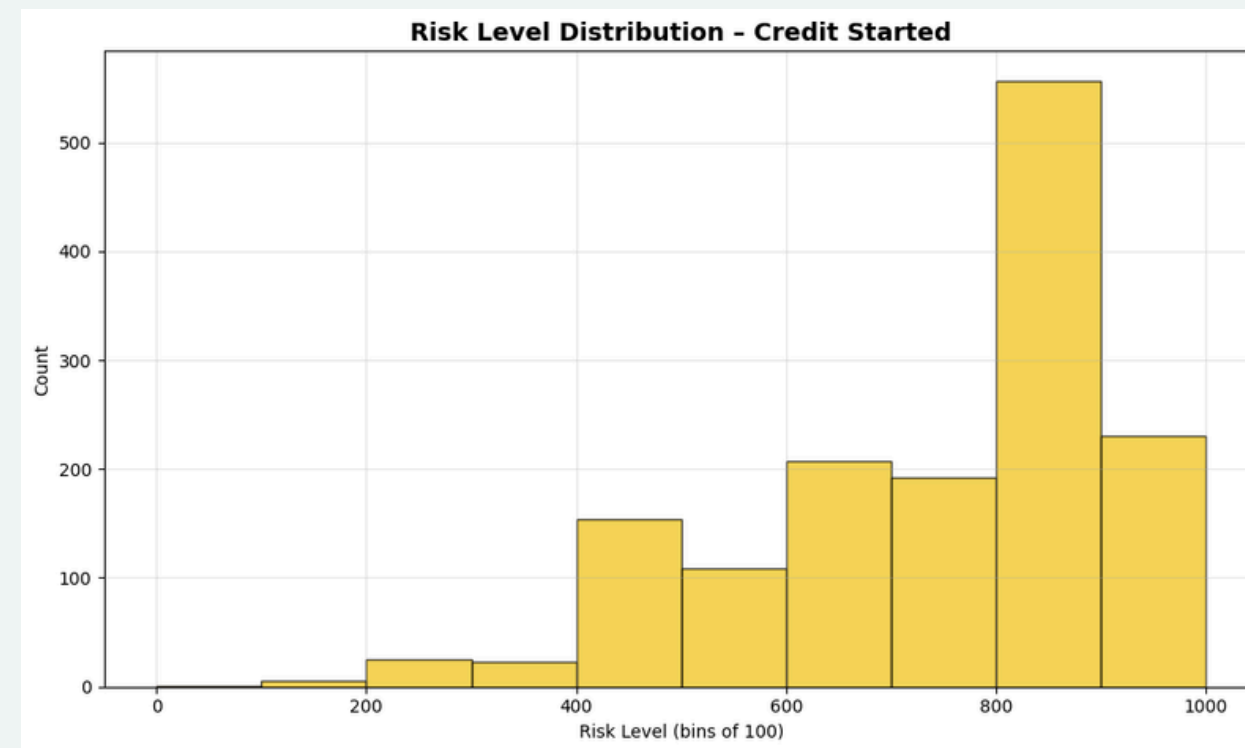
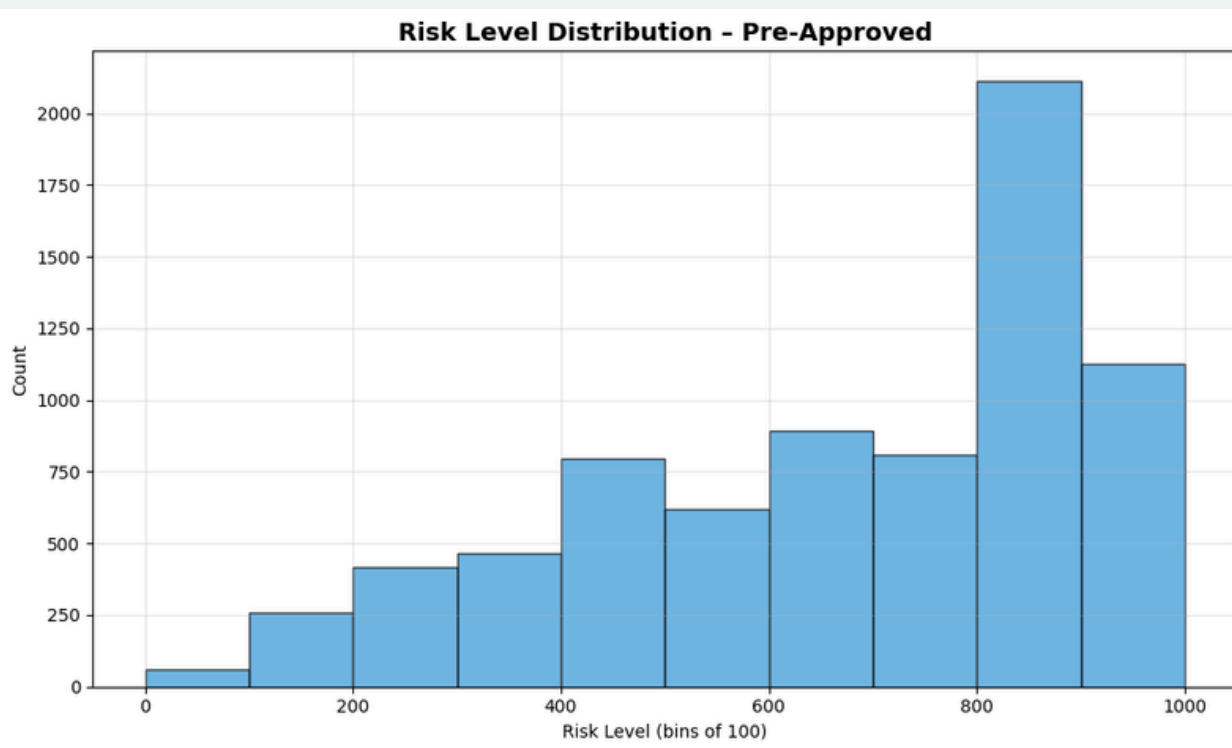
- All applicants peak in October–November, showing strong seasonal demand late in the year and also the increase of the marketing campaigning
- Credit Started grows slowly and peaks in November, but stays far below total applicants.
- Sales (Invoice) peak around September–October.
- There's a big drop-off between stages, meaning only a small fraction of applicants reach the final sale.
- December has a few values but we don't consider for the analysis

QUESTION 8: LEVEL OF APPROVAL BY SUPERVISOR OF SALES FROM CONVERSIONS (RESPONSIBILITY)



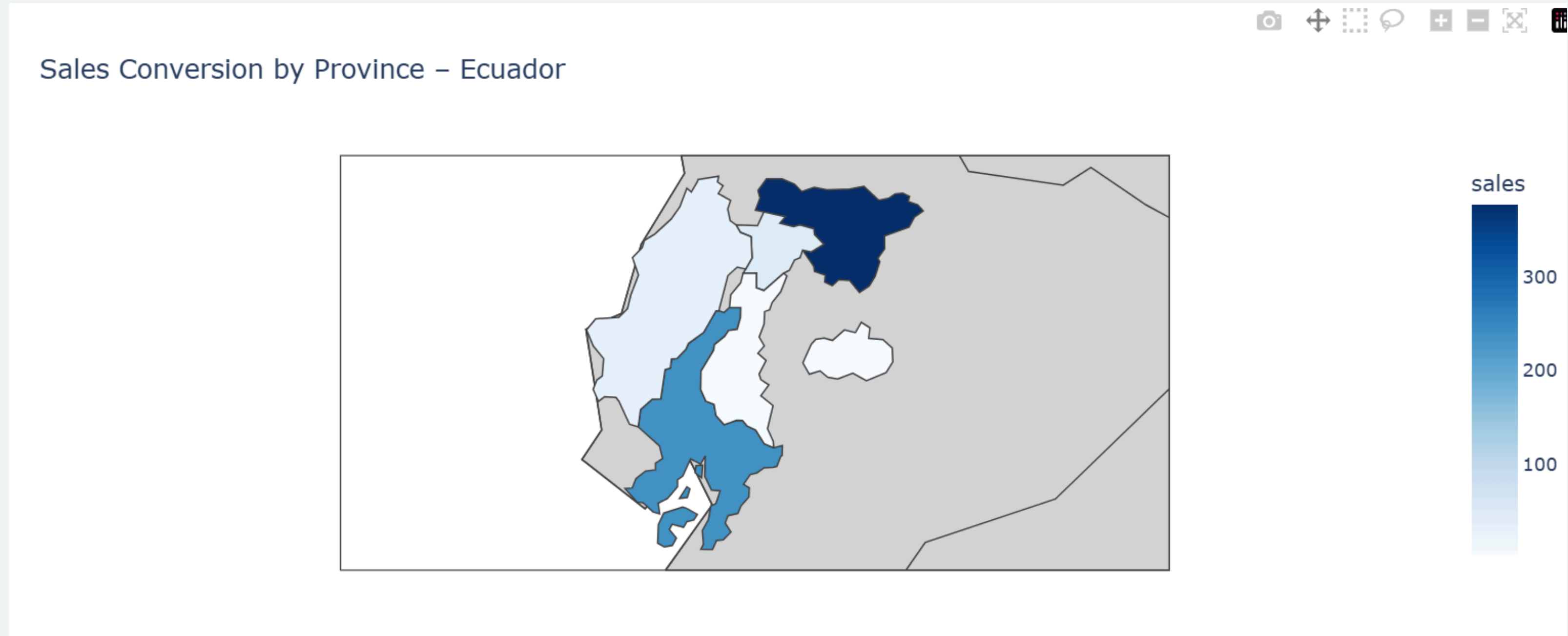
- The Analyst category dominates the final sales stage, meaning most successful invoices come from cases handled at this level. This suggests analysts are either receiving the strongest applicants or applying criteria that lead to higher conversion.
- We need to see the number of stores in charge to all supervisors and see what is best

QUESTION 9: SCORING HISTOGRAM BY STAGE?



- The biggest concentration is in the 800–900 bin, meaning the majority of pre-approved people have low-risk scores.
- The 900–1000 score is the next biggest, reinforcing that pre-approval.
- Low and medium risk levels are almost nonexistent. Scores below 700 barely appear, showing that very few low-risk clients reach pre-approval.
- The next step is to be aware of the default people

QUESTION 10: MAP OF THE SALES TO VISUALIZE THE REGIONS TO IMPROVE THE SALES?



- Guayas and Pichincha dominate credit sales; these two provinces clearly generate the highest conversion rates.
- Manabí, El Oro, and Santa Elena show a stronger conversion rate than many Andean provinces.
- We can shift our strategy toward the coastal region to boost these sales through marketing and by leasing new locations.

CONCLUSIONS

1. Approval Flow:

- The conversion is 2.7% its an average value, but the strategy is to increase awareness through marketing at least double by the month.
- About 54% of applicants are approved, but only a small portion start the credit process. We propose to trade off at least the 60% get the risk but increase the revenue.

2. Best-Performing Segment: Formal employees and younger applicants show the highest conversion rates. Increase the marketing campaign in this segment.

3. Weak Segments: Older applicants and informal/self-employed workers rarely reach the final sale stage.

4. Financial Profile: Most clients have low income and low expenses, matching a high-risk, small-ticket credit model. Control the default of the people from the first month.

5. Geographic Patterns: Guayas and Pichincha lead in both pre-approvals and sales; coastal provinces show strong growth potential. We propose to rent more stores in Guayas, at least 1 more.

6. Risk Scores: Most approved clients fall in the 800–900 risk range, indicating a predominantly low-risk portfolio. But we need to control the default payment.